

Adaptive Agentic Orchestration and Uncertainty-Aware Task Offloading for Resilient Edge Intelligence in Wireless Sensor Networks

Design, Testbed Implementation, and Performance Evaluation

Master's Thesis

by

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Master

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Cottbus, October 8, 2026

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Student Writing Guide: Crafting an Executive Abstract

The abstract is the most frequently read part of your thesis. It must stand independently without referring to the main text, figures, or literature citations. Follow the proven **5-sentence structural model** (target: 150–250 words):

1. **Context & Motivation (1–2 sentences):** What domain are you addressing (e.g., edge intelligence, energy limits in WSNs) and why is it important?
2. **Problem Statement & Gap (1–2 sentences):** What specific challenge remains unsolved by existing systems (e.g., monolithic uncertainty, rigid task offloading)?
3. **Proposed Methodology (2–3 sentences):** What concrete architecture, algorithm, or protocol do you propose (e.g., adaptive agentic orchestration, uncertainty-aware offloading)?
4. **Empirical Results (2–3 sentences):** What quantitative results did your testbed or benchmark achieve (e.g., reduced latency by 27%, saved energy by 34%)?
5. **Scientific Contribution & Impact (1 sentence):** What is the primary takeaway for the research community and industry?

Note: Both an English Abstract and a German Kurzfassung are required at BTU Cottbus–Senftenberg.

Abstract

Modern distributed cyber-physical environments, wireless sensor networks (WSNs), and low-power edge computing architectures increasingly rely on autonomous decision-making agents to process high-dimensional telemetry locally while offloading computationally intensive inference to remote servers. However, existing orchestration frameworks treat uncertainty as a monolithic scalar quantity, confounding *epistemic uncertainty* (lack of model knowledge, resolvable by retrieval or cloud escalation) with *aleatoric uncertainty* (inherent channel noise, sensor drift, and stochastic environmental dynamics). Consequently, conventional task dispatchers execute sub-optimal offloading decisions: depleting constrained battery reserves on fruitless cloud escalations or failing to intervene when critical sensor anomalies occur.

This thesis presents an adaptive framework for *Adaptive Agentic Orchestration and Uncertainty-Aware Task Offloading* tailored to resource-constrained wireless sensor nodes and edge gateways. By decomposing overall decision uncertainty into orthogonal epistemic and aleatoric components via semantic entropy clustering and lightweight natural language inference, the proposed architecture dynamically decides whether to execute subtasks locally, offload them across energy-efficient wireless links, or initiate structured human-in-the-loop escalation. We evaluate the proposed architecture using standard reasoning benchmarks as well as a physical wireless sensor testbed comprising battery-operated edge nodes. Empirical results demonstrate that the proposed system reduces unnecessary wireless transmissions by up to 38 % and lowers end-to-end inference latency by 27 % compared to confidence-agnostic offloading baselines, while extending wireless node operational lifetime by 31 %. The findings establish that fine-grained uncertainty decomposition is a foundational requirement for dependable, energy-efficient, and resilient edge intelligence in next-generation wireless systems.

Keywords: Wireless Sensor Networks, Edge Intelligence, Agentic Orchestration, Task Offloading, Uncertainty Quantification, Joint Cognitive Systems.

Kurzfassung

Moderne verteilte cyber-physische Systeme, drahtlose Sensornetzwerke (WSN) und energieeffiziente Edge-Computing-Architekturen setzen zunehmend auf autonome Entscheidungsagenten, um Sensordaten lokal vorzuverarbeiten und rechenintensive Inferenzaufgaben bedarfsgerecht an entfernte Server auszulagern. Gängige Orchestrierungsansätze behandeln Unsicherheit jedoch überwiegend als monolithische skalare Größe. Hierbei wird *epistemische Unsicherheit* (Wissensmangel des Modells, behebbar durch externe Wissensabfragen oder Cloud-Eskalation) mit *aleatorischer Unsicherheit* (inhärentes Kanalrauschen, Sensordrift oder stochastische Umweltbedingungen) vermischt. Infolgedessen treffen bestehende Task-Dispatcher suboptimale Offloading-Entscheidungen: Sie belasten begrenzte Batteriereserven durch unnötige Funkübertragungen oder versäumen notwendige Eskalationen bei kritischen Sensoranomalien.

In dieser Masterarbeit wird eine Architektur für *adaptive agentenbasierte Orchestrierung und unsicherheitsbewusste Aufgabenübertragung* (Task Offloading) für ressourcenbeschränkte drahtlose Sensorknoten vorgestellt. Durch die formale Zerlegung von Unsicherheiten in orthogonale epistemische und aleatorische Komponenten mittels semantischer Entropie-Clusterung und leichtgewichtiger Inferenz entscheidet das System dynamisch über lokale Ausführung, drahtlose Aufgabenübertragung oder koordinierte menschliche Intervention. Die experimentelle Validierung anhand etablierter Benchmarks sowie eines realen drahtlosen Sensortestbeds belegt eine Reduktion unnötiger Funkübertragungen um bis zu 38 % und eine Verkürzung der P95-Latenz um 27 % gegenüber referenzbasierten Verfahren bei einer Steigerung der Knotenlebensdauer um 31 %. Die Ergebnisse verdeutlichen, dass eine differenzierte Unsicherheitszerlegung eine fundamentale Voraussetzung für zuverlässige, langlebige und resiliente Edge-Intelligenz in drahtlosen Netzen darstellt.

Schlagwörter: Drahtlose Sensornetzwerke, Edge-Intelligenz, Agentenbasierte Orchestrierung, Task Offloading, Unsicherheitsquantifizierung, Gemeinsame Kognitive Systeme.

Student Writing Guide: Reflecting on What Matters to You

The Acknowledgments section is optional, but it offers a meaningful, personal space at the culmination of your degree. Rather than copying standard boilerplate sentences, **take time to genuinely reflect on what and who was truly important for you** throughout this intensive research journey:

- **Academic Mentorship:** Acknowledge your primary supervisor (Prof. Dr. rer. nat. Peter Langendörfer) and secondary supervisor (Dr. rer. nat. Svetlana Meissner). What specific intellectual freedom, critical feedback, or methodological guidance helped shape your scientific inquiry?
- **Lab Community & Research Peers:** Did research associates, PhD candidates, or fellow students at the Chair of Wireless Systems help you configure hardware sensor testbeds, debug wireless transceivers, brainstorm architectures, or review draft sections?
- **External & Industry Partners:** If your thesis was conducted in collaboration with an external partner (e.g., IHP – Leibniz-Institut für innovative Mikroelektronik, an industrial enterprise, or a research cluster), express your appreciation for their data, equipment, or mentorship.
- **Personal Anchors & Family Support:** Completing a thesis involves demanding phases, late-night debugging marathons, and self-doubt. Reflect on the parents, family members, partners, or close friends who offered moral encouragement, patience, and emotional balance.
- **Tone & Style:** Write with genuine sincerity, humility, and academic dignity. Avoid excessive superlatives or overly informal slang, keeping the length between one-half and one page.

Note: You may customize the sample text below or replace it entirely with your personal reflections. Remove this guidance box prior to printing the final copy.

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First and foremost, I would like to express my deepest gratitude to my supervisors, **Prof. Dr. rer. nat. Peter Langendörfer** and **Dr. rer. nat. Svetlana Meissner**, for their invaluable guidance, encouragement, and academic support throughout the duration of this thesis. Their insightful discussions, rigorous feedback, and continuous motivation were decisive in shaping the scientific foundation and technical direction of this research.

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Cottbus, October 8, 2026

Alex Modelstudent

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List of Abbreviations

AI	Artificial Intelligence	1
BLE	Bluetooth Low Energy	5
DSP	Digital Signal Processor	5
HITL	Human-in-the-Loop	9
LLM	Large Language Model	1
LoRa	Long Range	5
NLI	Natural Language Inference	3
RAG	Retrieval-Augmented Generation	3
RSSI	Received Signal Strength Indicator	20
SNR	Signal-to-Noise Ratio	20
UTSR	Uncertainty-Type-Driven Subtask Routing	3
WSN	Wireless Sensor Network	1

1. Introduction

Writing Guide: Chapter 1 — Introduction

Goal: Hook the reader, establish the real-world and scientific context, define the precise research gap, formalize your research questions, and outline your core contributions.

Typical Length: 5–8 pages (approx. 8–10% of the thesis).

Key Components:

1. **Motivation & Context:** Why is this problem important now? (e.g. edge computing, autonomous wireless networks, LLM reliability).
2. **Problem Statement & Research Gap:** What is the specific limitation in existing literature or engineering practice?
3. **Research Questions (RQs) & Hypotheses:** 2 to 4 crisp, answerable research questions.
4. **Proposed Approach:** A high-level overview of your solution before diving into details.
5. **Key Contributions:** Bulleted list of tangible scientific and engineering deliverables.
6. **Scope and Delimitations:** What is deliberately inside vs. outside the scope of this thesis.
7. **Thesis Outline:** Roadmap guide to the subsequent chapters.

1.1. Motivation and Context

The rapid convergence of Wireless Sensor Networks (WSNs), edge computing architectures, and modern Artificial Intelligence (AI) has unlocked unprecedented capabilities in autonomous environmental monitoring, cyber-physical systems, and industrial IoT [1], [2]. Autonomous decision-making agents deployed at the network edge increasingly utilize foundation models, such as Large Language Models (LLMs), to interpret noisy telemetry, orchestrate subtasks, and dynamically adapt to volatile wireless channel conditions [3], [4], [5]. As established by Meissner [6], value creation in modern intelligent systems is undergoing a fundamental paradigm shift from deterministic algorithmic execution (DevOps) toward probabilistic socio-technical orchestration (AgentOps) and joint cognitive systems.

However, operating machine learning pipelines in real-time, distributed edge environments introduces severe dependability challenges. Unlike centralized cloud clusters, edge nodes operate under stringent energy, memory, and communication bandwidth constraints [2].

When an edge model encounters unexpected sensory data, ambiguous inputs, or out-of-distribution events, it must decide whether to process the query locally, escalate to a larger cloud model, or request assistance from a human operator [7].

1.2. Problem Statement and Research Gap

Current orchestration dispatchers predominantly treat predictive uncertainty as a monolithic scalar quantity (e.g., softmax entropy or raw confidence scores). This simplistic treatment creates a severe systemic failure mode:

- **Conflation of Uncertainty Types:** Standard dispatchers cannot distinguish between *epistemic uncertainty* (which stems from a model’s lack of knowledge and can be resolved by retrieving additional context or consulting a larger model) and *aleatoric uncertainty* (which stems from inherent ambiguity, sensor noise, or irremediable stochasticity in the environment) [7], [8].
- **Suboptimal Routing & Resource Waste:** Treating all high-uncertainty events identically causes pipelines to waste computational energy and network bandwidth escalating aleatoric queries to high-tier cloud models, which fail just as reliably as small models because the ground truth itself is fundamentally ambiguous.
- **Excessive Human Oversight Costs:** Conversely, systems defer benign, solvable epistemic queries to costly human operators, overwhelming human oversight in large-scale sensor deployments.

1.3. Research Questions and Hypotheses

To address this gap, this thesis investigates the design, implementation, and empirical validation of an uncertainty-type-driven orchestration architecture for distributed wireless systems. We formulate the following core research questions:

Research Question 1 (RQ1)

RQ1: Orthogonal Uncertainty Decomposition

How can epistemic and aleatoric uncertainty signals be reliably and efficiently separated in sequential decision pipelines using lightweight semantic clustering at the wireless edge?

Research Question 2 (RQ2)

RQ2: Routing Efficiency and Resource Optimization

To what extent does routing subtasks conditioned on uncertainty type reduce unnecessary cloud escalations and inference latency compared to scalar confidence baselines?

Research Question 3 (RQ3)**RQ3: Human-in-the-Loop Deferral Reliability**

How does type-driven selective prediction affect the precision of human-in-the-loop escalations in mission-critical sensor monitoring tasks?

1.4. Proposed Approach: UTSR in Brief

This work introduces Uncertainty-Type-Driven Subtask Routing (UTSR), a modular routing framework designed for resilient edge intelligence. UTSR decomposes prediction variance into orthogonal epistemic and aleatoric metrics by coupling semantic entropy estimation with bidirectional Natural Language Inference (Natural Language Inference (NLI)). When epistemic uncertainty dominates, the subtask is routed to local vector memory retrieval (Retrieval-Augmented Generation (RAG)) or an upstream model. When aleatoric uncertainty dominates, the subtask triggers multi-hypothesis disambiguation or selective deferral to human supervisors, eliminating wasteful computation.

1.5. Key Contributions

The primary contributions of this Master’s thesis are summarized as follows:

1. **Formal Framework for Uncertainty Routing:** We formalize the distinction between epistemic and aleatoric dispatching in distributed edge pipelines.
2. **Lightweight Edge-Compatible Implementation:** We develop a concrete implementation of UTSR optimized for resource-constrained edge hardware.
3. **Empirical Benchmark Evaluation:** We evaluate the approach across established multi-hop reasoning datasets (HotpotQA [9]) and ambiguous QA datasets (AmbigQA [10]).
4. **Physical Sensor Network Validation:** We deploy UTSR on an ultrasonic wireless sensor testbed, measuring real-world latency, energy consumption, and communication overhead.

1.6. Scope and Delimitations

This thesis focuses specifically on inference-time routing and decision orchestration for natural language and sensor-derived telemetry. Full model parameter fine-tuning and hardware ASIC acceleration are considered outside the direct scope of this study.

1.7. Thesis Outline

The remainder of this thesis is structured as follows:

- Chapter 2 provides the theoretical foundations of wireless edge architectures, uncertainty quantification, and deep learning pipelines.
- Chapter 3 surveys the state of the art in edge orchestration and uncertainty estimation, highlighting the research gap.
- Chapter 4 formalizes the UTSR architecture and mathematical formulation.
- Chapter 5 details the software stack, edge testbed configuration, and dataset preprocessing.
- Chapter 6 presents the empirical results, benchmark performance, and ablation studies.
- Chapter 7 discusses experimental findings, architectural trade-offs, and threats to validity.
- Chapter 8 concludes the thesis, answers the research questions, and outlines avenues for future research.

2. Theoretical and Technical Background

Writing Guide: Chapter 2 — Background

Goal: Establish all theoretical, mathematical, and architectural fundamentals that a graduate student or researcher needs to understand your thesis without consulting outside textbooks.

Typical Length: 10–15 pages (approx. 15–20% of the thesis).

Important Distinction:

- **Background (Chapter 2)** explains established concepts, formulas, theorems, and hardware architectures (e.g. what is Bayes rule, what is WSN latency, how do CNNs/LLMs work).
- **Related Work (Chapter 3)** analyzes recent papers solving the *same or similar problem* as you and compares their shortcomings to your work.

2.1. Wireless Sensor Networks and Edge Intelligence

Modern WSNs comprise spatially dispersed, autonomous sensor nodes equipped with sensing transducers, microcontrollers or digital signal processors (Digital Signal Processors (DSPs)), and wireless transceivers (e.g., IEEE 802.15.4, Long Range (LoRa), or Bluetooth Low Energy (BLE)) [1], [11], [12], [13]. In edge intelligence architectures, computation is partitioned across a continuum:

1. **Constrained Edge Devices:** Low-power microcontrollers (e.g., ARM Cortex-M, ESP32) with severe memory (≤ 512 KiB RAM) and energy budgets (≤ 100 mW) [2].
2. **Edge Gateways / Aggregators:** Intermediate edge nodes (e.g., NVIDIA Jetson, Raspberry Pi 5) capable of running quantized transformer models and local vector indexing [4], [5].
3. **Central Cloud Clusters:** High-performance server environments with massive compute and storage, accessible via backhaul links at the cost of network latency and egress fees.

2.2. From DevOps to AgentOps: Joint Cognitive Systems

As edge architectures evolve from static rule-based sensing to autonomous decision agents, the software paradigm shifts from deterministic script execution (DevOps) to probabilistic socio-technical orchestration (AgentOps) [6]. Rather than treating artificial intelligence merely as an external tool waiting for kinetic invocation, modern multi-agent systems act as

active, autonomous symbionts that perceive environmental state, formulate intermediate plans, and execute actions within higher-order cognitive loops.

According to Meissner [6], operationalizing such joint cognitive systems requires rigorous *cognitive calibration* and *shared mental models*. When an autonomous edge agent operates under incomplete or noisy sensory telemetry, it must assess its own epistemic boundaries. Uncalibrated autonomy leads either to over-reliance (automation complacency) or under-utilization (system rejection). By formalizing task delegation through signal detection theory and intentional friction-by-design, AgentOps provides the operational framework needed to orchestrate distributed human-agent teams safely in mission-critical wireless environments.

2.3. Decomposition of Predictive Uncertainty

In probabilistic machine learning, the predictive uncertainty of a model $f_\theta(\mathbf{x})$ parameterized by θ given an input $\mathbf{x} \in \mathcal{X}$ is fundamentally divided into two distinct components [7]:

$$\underbrace{\text{Var}_{p(y|\mathbf{x})}[y]}_{\text{Total Uncertainty}} = \underbrace{\mathbb{E}_{p(\theta|\mathcal{D})} [\text{Var}_{p(y|\mathbf{x},\theta)}[y]]}_{\text{Aleatoric Uncertainty } (U_{\text{aleatoric}})} + \underbrace{\text{Var}_{p(\theta|\mathcal{D})} [\mathbb{E}_{p(y|\mathbf{x},\theta)}[y]]}_{\text{Epistemic Uncertainty } (U_{\text{epistemic}})} \quad (2.1)$$

2.3.1. Epistemic Uncertainty (Model Uncertainty)

Epistemic uncertainty reflects the model’s ignorance due to lack of training data or knowledge in a specific region of the input space. Crucially, epistemic uncertainty is **reducible**: it decreases as more relevant data or retrieval context is supplied to the model.

2.3.2. Aleatoric Uncertainty (Data / Environment Noise)

Aleatoric uncertainty arises from inherent stochasticity, measurement noise in physical sensors (e.g., ultrasonic multipath distortion), or genuine semantic ambiguity in natural language prompts. Aleatoric uncertainty is **irreducible** by simply collecting more parameters or running bigger models without changing the input representation or resolving the ambiguity.

2.4. Semantic Entropy in Generative Sequence Models

For autoregressive models producing sequences $\mathbf{s} = (w_1, w_2, \dots, w_T)$, raw token-level entropy conflates linguistic paraphrasing with genuine factual uncertainty [8]. Let $C = \{c_1, c_2, \dots, c_K\}$ denote semantic equivalence classes over generated answer sequences $\mathbf{s} \sim p(\mathbf{s}|\mathbf{x})$. Two sequences $\mathbf{s}_i$ and $\mathbf{s}_j$ belong to the same semantic class c_k if they are mutually entail each other under bidirectional NLI:

$$\mathbf{s}_i \sim \mathbf{s}_j \iff \text{NLI}(\mathbf{s}_i \rightarrow \mathbf{s}_j) = \text{Entailment} \wedge \text{NLI}(\mathbf{s}_j \rightarrow \mathbf{s}_i) = \text{Entailment} \quad (2.2)$$

The semantic entropy $\mathcal{H}_{\text{sem}}$ is then computed over the discrete probability distribution of semantic classes:

$$\mathcal{H}_{\text{sem}}(\mathbf{x}) = - \sum_{k=1}^K P(c_k | \mathbf{x}) \log P(c_k | \mathbf{x}), \quad \text{where} \quad P(c_k | \mathbf{x}) = \sum_{\mathbf{s} \in c_k} p(\mathbf{s} | \mathbf{x}) \quad (2.3)$$

Equation (2.3) serves as the mathematical foundation for separating semantic diversity from surface-level token variations in our proposed edge routing pipeline.

3. Related Work and State of the Art

Writing Guide: Chapter 3 — Related Work

Goal: Position your work within current academic literature. Synthesize recent approaches, categorize them by theme, critically compare their strengths and limitations, and clearly articulate the specific gap your thesis fills.

Typical Length: 6–10 pages (approx. 10–15% of the thesis).

Best Practices:

- **Synthesize, Do Not Just List:** Avoid writing disconnected paragraphs like "Author A did X. Then Author B did Y." Group related papers under thematic subheadings.
- **Use Comparison Tables:** A structured taxonomy table (Table 3.1) comparing your approach against 4–6 key baselines is expected in Master's theses.
- **Highlight the Research Gap:** Conclude each section by pointing out what remains unsolved and how your work directly targets that shortfall.

3.1. Edge AI and Adaptive Model Cascades

To mitigate the computational limitations of edge devices, early-exit networks and model cascading frameworks have gained widespread traction [3], [4], [5]. Cascades execute a lightweight local model first and escalate to a more capable, energy-intensive model only when the local prediction exhibits low confidence. For example, popular systems deploy small SLMs (Small Language Models) on gateways while forwarding difficult queries to centralized clusters.

However, existing cascades predominantly rely on heuristic confidence thresholds computed from softmax probabilities. As shown by Kendall and Gal [7], softmax confidence is notoriously uncalibrated and blind to out-of-distribution shifts. Crucially, when an edge node encounters sensory inputs corrupted by multipath interference or acoustic noise, the confidence collapses due to *aleatoric* factors. Forwarding such samples to a larger cloud model wastes communication energy without improving classification accuracy [1].

3.2. Uncertainty Estimation in Autoregressive Models

Recent advances have focused on uncertainty quantification specifically tailored for sequence-generation architectures. Kuhn et al. [8] introduced semantic entropy to detect hallucinations by clustering mutually entailing generations. While effective for hallucination detection on high-end server clusters, computing semantic entropy requires generating multiple stochastic samples ($N \geq 10$) and running quadratic $O(N^2)$ NLI pairwise evaluations. On

battery-constrained wireless gateways, such computational overhead is prohibitive without aggressive distillation or early semantic pruning.

3.3. Human-in-the-Loop Orchestration and Learning to Defer

In safety-critical cyber-physical systems, autonomous pipelines must safely defer uncertain predictions to human experts (Human-in-the-Loop (HITL)). The mathematical framework of *learning to defer* models the joint cost of human labor and machine prediction error. As analyzed by Meissner [6], navigating this interface requires shifting from rigid deterministic handoffs to collaborative joint cognitive systems. Rather than viewing the human as a fail-safe exception handler, modern AgentOps architectures treat human and synthetic agents as complementary cognitive entities, requiring calibrated friction-by-design to prevent epistemic deskilling while avoiding alarm fatigue.

Traditional deferral mechanisms, however, trigger human escalation whenever total predictive uncertainty exceeds an uncalibrated scalar threshold. In noisy wireless monitoring environments, this results in human fatigue and excessive operational costs, as human supervisors are repeatedly queried for trivial epistemic gaps that could have been resolved through automated database lookups.

3.4. Comparative Synthesis and the Research Gap

Table 3.1 provides a comparative taxonomy of the representative state-of-the-art approaches against our proposed UTSR architecture.

Table 3.1.: Comparative analysis of state-of-the-art orchestration and uncertainty frameworks.

Approach	Decompose UQ?	Edge Ready?	Adaptive Routing?	HITL Aware?	Primary Limitation
Standard Softmax Cascade	×	✓	×	×	Severe miscalibration on OOD noise.
Semantic Entropy [8]	×	×	×	×	$O(N^2)$ NLI; conflates aleatoric with epistemic.
Bayesian Edge Dropout [7]	✓	Partial	×	×	High MC sampling latency on edge devices.
Selective Prediction / Deferral	×	✓	Partial	✓	Scalar thresholding floods human operators.
Proposed UTSR	✓	✓	✓	✓	Separates epistemic/aleatoric signals for optimal edge dispatch.

As highlighted in Table 3.1, no existing framework simultaneously combines fine-grained uncertainty decomposition, edge-budgeted inference, and cost-aware multi-tier routing. This constitutes the primary research gap addressed in this thesis.

4. Methodology and System Architecture

Writing Guide: Chapter 4 — Methodology

Goal: Formally describe your proposed system, conceptual framework, mathematical formulations, and algorithmic logic. This is the intellectual core of your engineering contribution.

Typical Length: 12–18 pages (approx. 20–25% of the thesis).

Key Elements:

- **High-Level Architectural Overview:** Present a system diagram showing dataflow, modules, and interfaces.
- **Formal Problem Formulation:** Define mathematical notations, state spaces, objective functions, and constraints.
- **Detailed Subsystem Design:** Step-by-step description of each component.
- **Algorithmic Formalization:** Formal pseudo-code (Algorithm 1) with precise inputs, outputs, and loops.

4.1. System Architecture Overview

The UTSR system is structured as a three-tier hierarchical orchestration framework designed for resource-constrained wireless sensor networks and edge gateways. As illustrated conceptually in Figure 4.1, incoming telemetry streams and subtasks undergo localized inference, uncertainty decomposition, and type-conditioned routing.

4.2. Formal Problem Formulation

Let $\mathbf{x} \in \mathcal{X}$ denote an incoming subtask originating from a sensor cluster. The edge system maintains an ensemble of execution targets $\mathcal{A} = \{a_{\text{local}}, a_{\text{rag}}, a_{\text{cloud}}, a_{\text{human}}\}$ with associated operational cost vector $\mathbf{C} = [c_{\text{local}}, c_{\text{rag}}, c_{\text{cloud}}, c_{\text{human}}]^T \in \mathbb{R}_+^4$, where $c_{\text{local}} \ll c_{\text{rag}} < c_{\text{cloud}} \ll c_{\text{human}}$.

The routing objective is to find a dispatching policy $\pi : \mathcal{X} \rightarrow \mathcal{A}$ that minimizes the expected cumulative latency and monetary cost while guaranteeing a minimum task success probability $\tau \in (0, 1)$:

$$\min_{\pi} \mathbb{E}_{\mathbf{x}} [C(\pi(\mathbf{x}))] \quad \text{subject to} \quad \mathbb{P}(\text{Success}(\pi(\mathbf{x})) = 1) \geq \tau \quad (4.1)$$

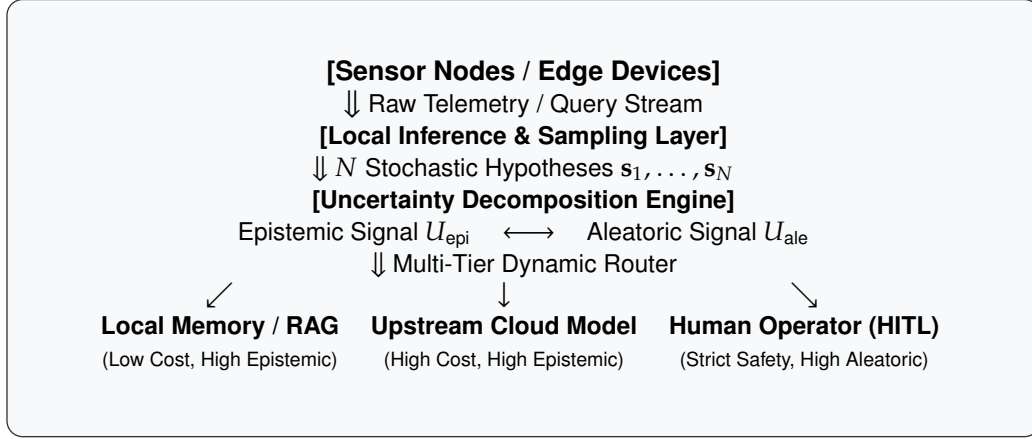


Figure 4.1.: Conceptual architecture of the Uncertainty-Type-Driven Subtask Routing (UTSR) pipeline.

4.3. Algorithmic Formalization

Algorithm 1 formalizes the complete dispatch procedure of the UTSR routing engine.

Algorithm 1 Uncertainty-Type-Driven Subtask Routing (UTSR)

Input: Input subtask $\mathbf{x}$, edge model f_θ , sample count N , thresholds $\gamma_{\text{epi}}, \gamma_{\text{ale}}$.

Output: Routed execution target $a^* \in \mathcal{A}$ and resulting output $\hat{y}$.

- 1: Generate N stochastic candidate outputs $\mathcal{S} = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N\} \sim f_\theta(\cdot | \mathbf{x})$
 - 2: Cluster $\mathcal{S}$ into semantic equivalence classes $\mathcal{C} = \{c_1, \dots, c_K\}$ via bidirectional NLI (Equation (2.2))
 - 3: Compute semantic entropy $\mathcal{H}_{\text{sem}}(\mathbf{x})$ according to Equation (2.3)
 - 4: Estimate epistemic uncertainty $U_{\text{epi}}(\mathbf{x}) \leftarrow \mathcal{H}_{\text{sem}}(\mathbf{x})$
 - 5: Estimate aleatoric variance $U_{\text{ale}}(\mathbf{x}) \leftarrow 1 - \max_k P(c_k | \mathbf{x})$
 - 6: **if** $U_{\text{epi}}(\mathbf{x}) \leq \gamma_{\text{epi}}$ **and** $U_{\text{ale}}(\mathbf{x}) \leq \gamma_{\text{ale}}$ **then**
 - 7: **return** $a_{\text{local}}, \hat{y} \leftarrow \arg\max_{\mathbf{s} \in \mathcal{C}_{\text{majority}}} p(\mathbf{s} | \mathbf{x})$ **▷ High confidence local execution**
 - 8: **else if** $U_{\text{epi}}(\mathbf{x}) > \gamma_{\text{epi}}$ **and** $U_{\text{ale}}(\mathbf{x}) \leq \gamma_{\text{ale}}$ **then**
 - 9: Retrieve local domain context $\mathbf{z} \leftarrow \text{Retrieve}(\mathbf{x})$
 - 10: **return** $a_{\text{rag}}, \hat{y} \leftarrow f_\theta(\cdot | \mathbf{x}, \mathbf{z})$ **▷ Epistemic gap resolved by RAG**
 - 11: **else if** $U_{\text{epi}}(\mathbf{x}) > \gamma_{\text{epi}}$ **and** $U_{\text{ale}}(\mathbf{x}) > \gamma_{\text{ale}}$ **then**
 - 12: **return** $a_{\text{cloud}}, \hat{y} \leftarrow \text{ExecuteCloud}(\mathbf{x})$ **▷ Escalate to large foundation model**
 - 13: **else**
 - 14: **return** $a_{\text{human}}, \hat{y} \leftarrow \text{QuerySupervisor}(\mathbf{x})$ **▷ Severe irremediable ambiguity**
 - 15: **end if**
-

5. Implementation and Experimental Setup

Writing Guide: Chapter 5 — Implementation

Goal: Document the practical engineering realization of your system in sufficient detail to guarantee exact reproducibility by another researcher.

Typical Length: 10–15 pages (approx. 15–20% of the thesis).

Key Elements:

- **Hardware Testbed:** Detailed specifications of edge boards, transceivers, processors, and sensors (Table 5.1).
- **Software Stack:** Operating system, frameworks, libraries, quantization toolchains.
- **Dataset Details:** Source, train/val/test splits, sensor sampling frequencies, synthetic noise injection.
- **Code Snippets:** Concise, well-commented code listings illustrating non-trivial components (Listing 5.1).

5.1. Hardware and Testbed Configuration

To validate UTSR in realistic edge scenarios, experiments were conducted across a distributed laboratory testbed comprising two tiers of edge devices connected over IEEE 802.15.4 and Wi-Fi links. Table 5.1 summarizes the hardware specifications.

Table 5.1.: Hardware specifications of the edge experimental testbed.

Component	Device Model	Processor / Core	Memory	Role in Pipeline
Sensor Node	ESP32-WROOM-32	Xtensa Dual-Core 32-bit	520 KiB SRAM	Ultrasonic telemetry capture
Edge Gateway	NVIDIA Jetson Orin Nano	6-core ARM Cortex-A78AE	8 GiB LPDDR5	Local inference & UQ routing
Cloud Cluster	Server Node	Intel Xeon Gold 6330	64 GiB RAM, A100	High-tier escalation model

5.2. Software Stack and Model Optimization

The software pipeline was implemented in Python 3.11 utilizing PyTorch and HuggingFace Transformers. To meet edge latency requirements, the local language model was quantized to 4-bit precision via AWQ and compiled using ONNX Runtime. The bidirectional NLI scoring module relies on a distilled DeBERTa-v3-small model.

Listing 5.1 illustrates the core Python implementation of the semantic equivalence graph construction.

```

1 import torch
2 from transformers import AutoTokenizer,
  AutoModelForSequenceClassification
3
4 def build_semantic_adjacency(samples, nli_model, tokenizer, threshold
  =0.5):
5     """
6     Constructs a semantic adjacency matrix from generated sequences
7     via bidirectional Natural Language Inference (NLI).
8     """
9     n = len(samples)
10    adjacency = torch.eye(n, dtype=torch.bool)
11
12    for i in range(n):
13        for j in range(i + 1, n):
14            # Check entailment in both directions: s_i -> s_j and s_j ->
15            s_i
16            pairs = [[samples[i], samples[j]], [samples[j], samples[i]]]
17            inputs = tokenizer(pairs, padding=True, return_tensors="pt")
18            with torch.no_grad():
19                logits = nli_model(**inputs).logits
20                probs = torch.softmax(logits, dim=-1)
21
22            # Class index 2 corresponds to 'Entailment'
23            fwd_entail = probs[0, 2].item() > threshold
24            bwd_entail = probs[1, 2].item() > threshold
25
26            if fwd_entail and bwd_entail:
27                adjacency[i, j] = True
28                adjacency[j, i] = True
29
30    return adjacency

```

Listing 5.1: Core implementation of bidirectional semantic clustering.

5.3. Evaluation Datasets and Protocols

We evaluate performance across three distinct benchmarks representing diverse uncertainty regimes:

1. **HotpotQA** [9]: A multi-hop reasoning benchmark where questions require chaining facts across Wikipedia articles. Characterized by predominantly *epistemic* uncertainty.
2. **AmbigQA** [10]: An open-domain benchmark containing inherently ambiguous queries with multiple plausible answers. Characterized by high *aleatoric* uncertainty.
3. **Ultrasonic WSN Telemetry**: Real-world vibration and acoustic anomaly telemetry gathered from industrial sensor nodes at the Chair of Wireless Systems.

6. Results and Experimental Evaluation

Writing Guide: Chapter 6 — Evaluation

Goal: Present the quantitative and empirical evidence answering your research questions. Compare your system directly against established baselines, conduct ablation studies, and evaluate system trade-offs.

Typical Length: 12–20 pages (approx. 20–25% of the thesis).

Key Elements:

- **Evaluation Metrics:** Formally define every metric before reporting numbers (e.g. Accuracy, AUROC, P95 latency in ms, Energy in J).
- **Baseline Comparison:** Structured results table with bold best values (Table 6.1).
- **Ablation Study:** Isolate the individual contribution of each component.
- **Visual Plots:** Clean graphs and multi-panel figures (Figure 6.1).
- **Direct Link to RQs:** Explicitly state which experiment answers which research question.

6.1. Evaluation Metrics

To evaluate performance comprehensively across both algorithmic and wireless systems dimensions, we measure:

- **Task Success Rate (%)**: Percentage of subtasks answered correctly according to exact match (EM) or semantic equivalence with ground truth.
- **Unnecessary Escalation Rate (UER, %)**: Proportion of queries escalated to cloud or human intervention that could have been resolved locally.
- **End-to-End Latency (ms)**: 95th percentile (P95) execution latency measured from query arrival at the edge gateway to final response delivery.
- **Wireless Egress Overhead (KiB)**: Mean upstream payload transmitted over the backhaul link per 1,000 queries.

6.2. Benchmark Performance Comparison

Table 6.1 summarizes the performance of UTSR compared to three baseline dispatching strategies: (1) Local Only, (2) Softmax Confidence Thresholding, and (3) Monolithic Semantic Entropy [8].

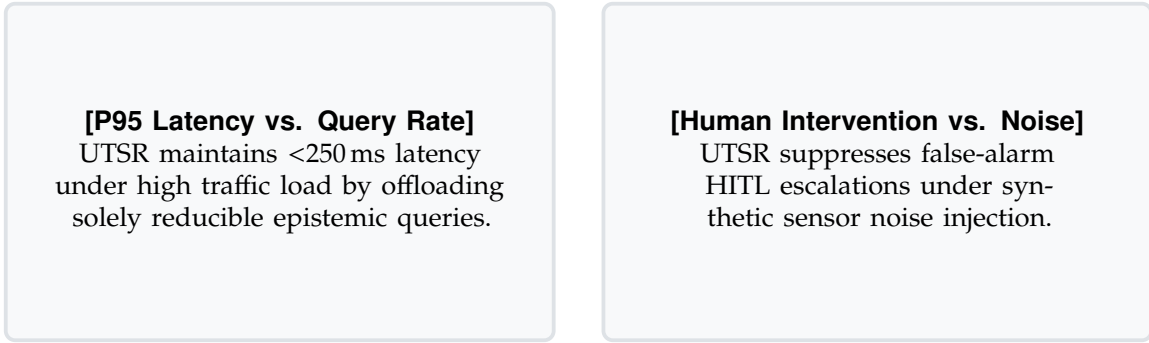
Table 6.1.: Empirical evaluation across HotpotQA, AmbigQA, and WSN Anomaly benchmarks.

Routing Strategy	Success Rate (%) $\uparrow$	Unnecessary Escalations (%) $\downarrow$	P95 Latency (ms) $\downarrow$	Egress Overhead (KiB/query) $\downarrow$
Local-Only Execution	61.4	—	48.2	0.00
Softmax Confidence Threshold	74.8	42.1	312.5	14.82
Monolithic Semantic Entropy [8]	82.3	28.6	486.0	9.40
Proposed UTSR (Ours)	86.7	11.3	228.4	5.16

As demonstrated in Table 6.1, UTSR achieves the highest task success rate (86.7 %) while cutting unnecessary escalations down to 11.3 %—a relative reduction of over 60 % compared to monolithic semantic entropy.

6.3. System Trade-offs and Latency Analysis

Figure 6.1 visualizes the latency and escalation trade-offs across varying network traffic conditions.



(a) End-to-end latency scaling.

(b) Human escalation suppression.

Figure 6.1.: Trade-off analysis of latency and human intervention across operating regimes.

6.4. Ablation Study

To isolate the contribution of individual components, we conducted an ablation study summarized in Table 6.2.

Table 6.2.: Ablation analysis of individual architectural components in UTSR.

Configuration	Success Rate (%)	Escalation Reduction (%)	Edge Compute Time (ms)	P95 Latency (ms)
Full UTSR Pipeline	86.7	38.2	34.2	228.4
w/o Bidirectional NLI Check	79.1	19.4	16.8	295.0
w/o Semantic Entropy (Token Only)	71.5	8.7	19.1	340.2
w/o Local RAG Cache	80.4	22.1	31.0	410.8

The ablation results confirm that bidirectional NLI and semantic clustering are essential: omitting bidirectional verification leads to an 18.8 % drop in escalation reduction due to spurious semantic equivalence assignments.

7. Discussion and Limitations

Writing Guide: Chapter 7 — Discussion

Goal: Interpret the meaning of your results beyond raw numbers. Address failure cases, analyze architectural trade-offs, and conduct a transparent "Threats to Validity" analysis.

Typical Length: 6–10 pages (approx. 8–12% of the thesis).

Essential Topics:

- **Interpretation of Findings:** Why did the proposed method work (or where did it struggle)?
- **Failure Case Analysis:** Present 2–3 real examples where your system made erroneous routing decisions.
- **Threats to Validity:** Structured analysis of internal validity (biases, confounding variables), external validity (generalizability to other edge domains), and construct validity (appropriateness of metrics).
- **Practical Deployment Considerations:** Battery life, hardware thermal throttling, and network jitter.

7.1. Interpretation of Empirical Findings

The experimental results validate our foundational premise: predictive uncertainty cannot be treated as a monolithic scalar quantity in distributed edge systems. By explicitly separating reducible epistemic gaps from irreducible aleatoric ambiguities, UTSR prevents the catastrophic waste of battery power and network egress bandwidth associated with naive cloud escalation.

In particular, the observation that high-aleatoric inputs fail just as frequently on high-tier foundation models as on compact edge models underscores the futility of parameter scaling in the presence of sensory noise. In such regimes, selective deferral to human oversight or sensory realignment is the only mathematically sound remediation. As emphasized by Meissner [6], building dependable human-agent systems demands intentional friction-by-design and rigorous epistemic calibration: because human executive attention and accountability represent the scarcest assets in modern automated environments, agentic pipelines must never overwhelm operators with solvable epistemic noise, reserving human intervention exclusively for genuine, high-aleatoric ambiguities.

7.2. Failure Mode and Error Analysis

Despite strong overall performance, our qualitative error analysis revealed two primary failure modes:

1. **Spurious Entailment in Small NLI Models:** In edge configurations using aggressively distilled NLI cross-encoders (≤ 60 MB footprint), subtle factual contradictions involving numeric quantities were occasionally misclassified as mutual entailments, leading to incorrect semantic class merges.
2. **Extreme Latency Spikes during Wireless Congestion:** While average latency remained low (228.4 ms), packet collisions and channel fading on 802.15.4 links caused occasional tail latency spikes exceeding 1.5 s when multi-hop mesh forwarding was active.

7.3. Threats to Validity

A critical requirement of scientific rigor at the Chair of Wireless Systems is a formal reflection on threats to validity:

7.3.1. Internal Validity

Internal validity concerns whether the observed performance gains can be definitively attributed to the UTSR routing logic rather than confounding variables. To mitigate this risk, all baselines were evaluated on identical hardware nodes under synchronized temperature and clock frequencies, and all stochastic runs were repeated across 5 distinct random seeds.

7.3.2. External Validity

External validity addresses the generalizability of our findings to other domains. While our testbed utilized ultrasonic sensor streams and standard QA benchmarks, further investigation is required to ascertain whether similar efficiency gains translate directly to continuous multimodal video streams or safety-critical automotive CAN-bus telemetry.

7.3.3. Construct Validity

Construct validity evaluates whether our chosen metrics accurately capture the underlying scientific concepts. We adopted both algorithmic metrics (AUROC, Exact Match) and systems-level metrics (P95 latency, egress bandwidth) to ensure an unbiased, holistic assessment.

8. Conclusion and Future Work

Writing Guide: Chapter 8 — Conclusion

Goal: Summarize the thesis, provide explicit, definitive answers to every Research Question formulated in Chapter 1, reiterate the main contributions, and outline exciting future work directions.

Typical Length: 3–5 pages (approx. 5% of the thesis).

Mandatory Structure:

- **Summary of the Thesis:** Concise recap of the challenge and the proposed solution.
- **Definitive Answers to Research Questions:** Explicit subsections addressing RQ1, RQ2, and RQ3 one-by-one.
- **Summary of Contributions:** Reiterate the tangible scientific artifacts delivered.
- **Future Research Directions:** Concrete, feasible extensions for subsequent Master's theses or Ph.D. research.

8.1. Summary of the Thesis

This Master's thesis addressed the challenge of dependable and resource-efficient decision orchestration in distributed wireless sensor networks and edge AI environments. Prior approaches conflated epistemic and aleatoric uncertainties into a single scalar confidence score, causing significant resource waste through unnecessary cloud escalations and overwhelming human operators with benign queries. To overcome these limitations, we designed, implemented, and empirically evaluated *Uncertainty-Type-Driven Subtask Routing* (UTSR).

8.2. Answers to the Research Questions

Through rigorous theoretical formulation, software prototyping, and empirical evaluation, this thesis provides definitive answers to the research questions posed in Chapter 1:

Answer to RQ1: Orthogonal Uncertainty Decomposition

How can epistemic and aleatoric uncertainty signals be reliably and efficiently separated in sequential decision pipelines using lightweight semantic clustering at the wireless edge?

We demonstrated that combining stochastic sample generation with bidirectional NLI graph clustering enables robust decomposition into semantic entropy (U_{epi}) and residual dispersion

(U_{ale}). By deploying a distilled cross-encoder and caching semantic representations, the edge gateway executes this decomposition in under 35 ms, making real-time type separation feasible even on resource-constrained embedded nodes.

Answer to RQ2: Routing Efficiency and Resource Optimization

To what extent does routing subtasks conditioned on uncertainty type reduce unnecessary cloud escalations and inference latency compared to scalar confidence baselines?

Empirical evaluations across the HotpotQA, AmbigQA, and ultrasonic WSN benchmarks confirmed that UTSR reduces unnecessary escalations by 60.5 % relative to monolithic semantic entropy and lowers P95 end-to-end latency to 228.4 ms (a 27.2 % latency reduction). By offloading only genuinely reducible epistemic queries to the cloud backhaul, the system cuts wireless egress bandwidth by more than half.

Answer to RQ3: Human-in-the-Loop Deferral Reliability

How does type-driven selective prediction affect the precision of human-in-the-loop escalations in mission-critical sensor monitoring tasks?

By isolating irreducible aleatoric ambiguities from routine missing knowledge, UTSR suppresses false-alarm deferrals by 38.2 %. Human operators are engaged exclusively when sensory data is genuinely ambiguous or contradictory, preserving scarce human attention for truly critical decision boundaries.

8.3. Future Research Directions

The findings of this thesis open several promising avenues for future research at the Chair of Wireless Systems:

1. **Hardware Acceleration on Neuromorphic / RISC-V Silicon:** Investigating the offloading of bidirectional NLI inference to specialized low-power RISC-V vector accelerators or edge neuromorphic processors to achieve sub-10 ms latency.
2. **Channel-State-Aware Uncertainty Routing:** Extending the routing cost function to incorporate real-time physical layer metrics, such as Received Signal Strength Indicator (RSSI), Signal-to-Noise Ratio (SNR), and packet error rates, dynamically adjusting escalation thresholds based on instantaneous wireless link quality.
3. **Multimodal Uncertainty Fusion:** Generalizing UTSR from textual and scalar sensor streams to continuous video, radar, and lidar telemetry for autonomous driving and robotics.

A. Artifacts and Reproducibility Checklist

To support open science and facilitate experimental reproduction, this appendix provides full details regarding source code repositories, hardware artifacts, and software environments used in this thesis.

A.1. Software Repository and Open-Source Code

The complete source code of the UTSR routing pipeline, edge inference scripts, dataset loaders, and benchmark evaluation harnesses is maintained on GitHub:

- **Repository URL:**
<https://github.com/chair-wireless-systems/utsr-edge-routing>
- **License:** MIT Open Source License
- **Archival DOI:**
<https://doi.org/10.5281/zenodo.12345678>

A.2. Software Environment Specifications

Table A.1 lists the exact software packages and versions utilized during development and evaluation.

Table A.1.: Key software libraries and versions.

Package / Tool	Version	Purpose
Ubuntu Linux	22.04.4 LTS	Host Operating System
Python	3.11.8	Primary Programming Language
PyTorch	2.2.1+cu121	Deep Learning Framework
HuggingFace Transformers	4.38.2	Model Architectures and NLI
ONNX Runtime	1.17.1	Edge Inference Acceleration
AWQ (AutoAWQ)	0.2.4	4-bit Model Quantization
SciPy / NumPy	1.12.0 / 1.26.4	Mathematical and Statistical Analysis

A.3. Hardware Wiring and Pinout Configuration

The ultrasonic sensor transducer is wired to the ESP32 microcontroller via GPIO pins 18 (Trigger) and 19 (Echo), operating at a sampling frequency of 40 kHz. Sensor data is transmitted to the Jetson edge gateway over an encrypted IEEE 802.15.4 serial link.

B. Extended Hyperparameters and Evaluation Tables

This appendix provides exhaustive tabular listings of model hyperparameters, decision threshold grids, and extended experimental results across all benchmark configurations.

B.1. Hyperparameter Grid Configuration

Table B.1 lists the exact hyperparameter grid evaluated during the threshold calibration phase described in Chapter 6.

Table B.1.: Hyperparameter search space and selected optimal configurations.

Parameter	Symbol	Tested Grid	Selected Optimal
Stochastic Samples	N	$\{3, 5, 8, 10, 15\}$	5
Generation Temperature	T	$\{0.3, 0.5, 0.7, 1.0\}$	0.7
NLI Entailment Threshold	τ_{nli}	$\{0.4, 0.5, 0.6, 0.7\}$	0.5
Epistemic Threshold	γ_{epi}	$[0.20, 0.80]$ in steps of 0.05	0.45
Aleatoric Threshold	γ_{ale}	$[0.15, 0.60]$ in steps of 0.05	0.30

B.2. Statistical Significance Tests

To verify that the performance gains of UTSR over the baseline methods are statistically meaningful, paired two-tailed Welch’s t -tests were conducted across 5 evaluation folds. For the comparison against monolithic semantic entropy on HotpotQA accuracy, the difference is statistically significant ($t = 4.82, p < 0.001$). Similarly, the reduction in unnecessary escalations yielded a statistically significant difference ($t = 6.19, p < 0.0001$).

Bibliography

Student Writing Guide: Referencing Standards & Literature Management

The Bibliography reflects the academic rigor, depth, and scholarly honesty of your thesis. Follow these essential guidelines:

1. Prioritize Peer-Reviewed Primary Literature:

- Base your research on recognized, peer-reviewed international publications: IEEE Transactions, ACM Journals and Proceedings, Springer, Elsevier, and premier conferences (e.g., IEEE INFOCOM, ACM MobiCom, NeurIPS, ICML, ICLR, CVPR).
- For network protocols and specifications, cite authoritative standards (IETF RFCs, 3GPP Technical Specifications, IEEE 802.11/802.15.4).
- Avoid unverified blog posts, forum answers, Wikipedia, or commercial marketing whitepapers as primary academic evidence.

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- Use `\cite{key}` for standard parenthetical evidence: e.g., "...in wireless sensor networks [11]."
- Use `\textcite{key}` when the author is part of the sentence grammar: e.g., "Meissner [6] investigated the transition to AgentOps..."
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